

Dual ordinal spectral ranking of journals and institutions

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Abstract

We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from intra-field citation flows. A continuous parameter χ apportions each reference between source and institution channels; a binary block-mask selects among four bibliometrically distinct flow structures, nesting conventional single-mode rankings as special cases. Applied to the economics and commerce literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: Scientometrics, spectral ranking, dual journal–institution prestige, Katz resolvent, Pinski–Narin

1 Introduction

Journal and institution ranking. The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval. Institutions fund individual researchers, implement training, oversee compliance, and maintain physical and information infrastructure. [Placeholder: overview of the joint ranking problem.]

Previous work on journal ranking. ? introduced influence-per-reference weights for journal ranking; Vigna (2016) surveys the broader lineage; ? traces the Katz–Hubbell–Pinski–Narin–PageRank chain. [Placeholder: expand journal ranking literature.]

How institutions are ranked. Institutional rankings, dominated by QS, Times Higher Education (THE), and ShanghaiRanking (ARWU), blend bibliometrics with reputational surveys, teaching metrics, and internationalisation. The Leiden Ranking counters with field-normalised bibliometrics, revealing niche leaders but lacking the reputational component of QS and THE. Spectral methods have not previously been applied jointly to sources and institutions. [Placeholder: expand institutional ranking literature.]

Theoretical framing. Following Wouters (1999), a *reference* written by a scientist becomes a *citation*—a directed edge (w_i, w_j) —only after indexer resolution to canonical work records.

We treat citations as structural relations within a functionally differentiated scientific system (?), setting aside authorial intent. Dual spectral rankings of journals and institutions treat citations as autopoietic communications within science’s communicative system (Social System of Science Citation Theory, SSCT), where individual actors reside in the environment rather than constituting system elements (Tahamtan and Bornmann, 2022). This contrasts with actor-network theory (ANT), which foregrounds heterogeneous agents in assembling citation networks; ANT enters only in the Discussion. The resulting dual rankings are self-referential hierarchies that condition future citations. Our construction extends Breiger (1974): all four blocks of the joint attention matrix are primary flows, not derived projections.

Paper structure. Section 2 develops the mathematical framework. Section ?? describes the corpus construction and source selection. Section 4 presents results. Section 5 discusses validity and prospects.

2 Journal–institution citation networks

We begin corpus construction by selecting N_s sources $\mathfrak{S}$ in a field, discipline or category. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $t_c = [t_0^c \dots t_1^c]$ or target years $t_t = [t_0^t \dots t_1^t]$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of authors $\mathfrak{A}'$ and institutions $\mathfrak{U}'$. Each work w_i has an *institution weight* under author-fractional counting (Waltman and van Eck, 2015),

$$\omega_{iu} = \sum_{\ell: u \in \Gamma_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, \quad (1)$$

where ℓ indexes authors of work w_i , a_i is the number of authors, and $u_{i\ell}$ the number of institutions of author ℓ . Using the work-institution frequency distribution to determine a minimum cut for works per institution per year $\geq \tau_U$, we retain the set $\mathfrak{U}$ and the corresponding set of authors $\mathfrak{A}$. Since ω_{iu} requires both an author and an institution, we drop any authorship where the institution $\notin \mathfrak{U}$, and subsequently drop works that lose all their authorships. The institution weights are not renormalised to account for the omission of an authorship, so that $\sum_u \omega_{iu} \leq 1$.

When a work w_i references another work w_j we say that w_i gives attention to w_j . The attention given by a work can be either (a) the number of references, or (b) a fixed amount (Zitt and Small, 2008). Let $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. We use the *reference normalisation weight* δ_i to select case (a) or case (b) respectively:

$$\delta_i = \begin{cases} 1 & \text{(full count)} \\ \bar{R}/R_i & \text{(fixed count)}. \end{cases} \quad (2)$$

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \dots, N_s$ and institutions indices $N_s + 1, \dots, N$. A reference $(i, j) \in \mathfrak{R}$ gives attention across unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (1). Attention flow between pairs of unit types is accumulated as four blocks over $\mathfrak{R}$ as rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \delta_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}, \quad (3)$$

with $\mathbf{b}_i^X$ weighting the referencing (citing) side and $\mathbf{b}_j^Y$ the referenced (cited) side.

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \delta) = \begin{pmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{pmatrix}, \quad (4)$$

where the binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$ is bibliometrically meaningful for source-only $(1, 0, 0, 0)$, institution-only $(0, 0, 0, 1)$, bipartite $(0, 1, 1, 0)$, and full-joint $(1, 1, 1, 1)$ combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The χ scaling preserves total citation flow across all block combinations, since $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$. We could name $\mathbf{C}$ the reference, attention or citation matrix and prefer the latter.

The model parameters are summarised in Table 1. Note that the corpus excludes (a) census work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

Spectral ranking. Let $r_p = \sum_q C_{pq}$ be the total outgoing citation flow from unit p , $\mathbf{D}_r = \text{diag}(r_p)$ the corresponding diagonal matrix. If $r_p = 0$ (a dangling unit with no outgoing citations) the corresponding row of $\mathbf{C}$ is zero and $\mathbf{H}$ is defined to have a zero row there; the boundary condition below then supplies the

Table 1: Model parameters θ .

Symbol	Name	Type	Role
<i>Corpus construction</i>			
$\mathfrak{S}$	source set	set	define the field, subject, category
$[t_0^c \dots t_1^c]$	census window	interval	publication years of works whose references are traced
$[t_0^t \dots t_1^t]$	target window	interval	publication years of works eligible to receive references
τ_U	institution threshold	integer	minimum work count to retain an institution
<i>Matrix construction</i>			
$\delta \in \{0, 1\}$	reference normalisation	binary	0: a work contributes $\bar{R}$ units 1: a work contributes R units
$\mathbf{m} \in \{0, 1\}^4$	block mask	binary 4-vector	select active unit-type pairs (SS , SI , IS , II)
$\chi \in [0, 1]$	mixing weight	continuous	source-institution balance within each work
<i>Ranking</i>			
$\alpha \in [0, 1)$	damping factor	continuous	controls propagation through citation network ensures convergence

necessary probability mass. Let $\boldsymbol{\mu}$ be an initial preference, or boundary condition (Hubbell, 1965; Vigna, 2016), with entries $\mu_p = a_p / \sum_q a_q$, where the unit work count a_p is $\Sigma_s = |\{i : w_i \in s\}|$ for source $p = s$ and $\Sigma_u = \sum_i \omega_{iu}$ for institution $p = u$, the latter being the fractional work count summed via (1). The row-stochastic citation matrix is $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$.

The spectral ranking $\boldsymbol{\pi}$ (see Vigna, 2016, for an overview) is defined as the solution to the damped eigenvector problem

$$\boldsymbol{\pi} = \alpha \mathbf{H}^\top \boldsymbol{\pi} + (1 - \alpha) \boldsymbol{\mu}, \quad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \quad (5)$$

with damping factor $\alpha \in [0, 1)$. The prior $\boldsymbol{\mu}$ encodes the distribution over units weighted by scholarly output, and α controls the degree to which influence propagates through the citation network rather than being absorbed by that prior. In the limit $\alpha \rightarrow 1$, equation (5) reduces to the principal eigenvector problem

$$\boldsymbol{\pi}^\infty = \mathbf{H}^\top \boldsymbol{\pi}^\infty, \quad \mathbf{1}^\top \boldsymbol{\pi}^\infty = 1, \quad \boldsymbol{\pi}^\infty \geq 0, \quad (6)$$

whose solution is unique when $\mathbf{H}$ is primitive (irreducible and aperiodic). Geller (1978) shows that $\boldsymbol{\pi}^\infty$ is then the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion $r_p / \sum_q r_q$.

The spectral ranking $\boldsymbol{\pi}$ reflects the total citation flow received by each unit and therefore depends on unit size. *Prestige per work* (Palacios-Huerta and Volij, 2004),

$$\mathbf{v}(\alpha) = \mathbf{A}^{-1} \boldsymbol{\pi}(\alpha), \quad \mathbf{A} = \text{diag}(a_p), \quad (7)$$

removes this dependence, where $\boldsymbol{\pi}(\alpha)$ denotes the solution to (5) and $\boldsymbol{\pi}^\infty$ the limiting case (6). Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004). Note that $\mathbf{A}^{-1}$ must be applied after convergence; applying it inside the iteration loop would iterate on $\mathbf{H}^\top \mathbf{A}$, which is not norm-preserving and yields a fixed point with no clean interpretation.

To compute $\boldsymbol{\pi}$ we define the modified matrix

$$\tilde{\mathbf{H}} = \alpha \mathbf{H} + (1 - \alpha) \boldsymbol{\mu} \mathbf{1}^\top, \quad (8)$$

so that the fixed-point condition $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$ is equivalent to equation (5). Because $\boldsymbol{\mu} > 0$ and $0 < \alpha < 1$, every entry of $\tilde{\mathbf{H}}$ is strictly positive, ensuring it is a primitive row-stochastic matrix. By the Perron–Frobenius theorem, $\tilde{\mathbf{H}}^\top$ therefore has a unique stationary distribution $\boldsymbol{\pi}$, recovered by power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \quad (9)$$

converging to π from any strictly positive starting vector $\pi^{(0)}$.

$\mathbf{C}(\chi, \mathbf{m}, \delta)$ is a sparse matrix built from a weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$ with per-reference weight δ_i , stored in CSR format, and row-normalised to $\mathbf{H}$. The pipeline $\mathfrak{R} \rightarrow \mathcal{E} \rightarrow \mathbf{C} \rightarrow \mathbf{H} \rightarrow \pi \rightarrow \mathbf{v}$ is implemented using `scipy.sparse.csr_matrix` with parameter entry points detailed in the Supplement.

3 Corpus selection

Corpus extraction starts with a list of eligible sources. We require each source to have three features: an OpenAlex source identifier for corpus extraction, evidence from one of our journal registries that the source publishes economics or business research, and relatively high OpenAlex source topic density to exclude cross-disciplinary journals and other edge cases.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is confined to Economics and Commerce so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having all FoR codes falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit ‘division’ codes and all four-digit ‘group’ codes (3501–3509, 3801–3803, 3899). After filtering, 1,602 journals remain. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance. After filtering, 1,413 journals remain.

Every journal in JQL, SJL, and MJL has at least one ISSN, and may have several. These registry ISSNs are joined independently to the OpenAlex sources snapshot using its list of ISSNs as the matching key. Journal titles are examined to identify around 30 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS*. The long list OAS* includes sources publishing works in disciplines beyond economics and business. We use OpenAlex source-topic assignments to exclude sources having a minority of works in economics/business. A source has work-topic counts for up to five topics, each within a single sub-field, field, and domain. We accumulate the work-topic counts for OpenAlex Field 14 (Economics, Econometrics and Finance) and Field 20 (Business and Management) and define the work-topic density is the ratio of this count to the total topic count in the source. Sources are dropped if this ratio is below 50%. The 1,659 surviving sources, labeled OAS, are used to extract the corpus. Matching outcomes are summarised in Table 2.

Table 2: Registry source matches to OpenAlex source identifiers

Register	Journals	Matched	Match rate	Unmatched
JQL (Harzing’s journal quality list)	842	839	99.6%	3
MJL (Clarivate master journal list)	1,413	1,392	98.5%	21
SJL (ERA submission journal list)	1,602	1,388	86.6%	214
Registries pairwise overlap				
MJL $\cap$ SJL				783 sources
MJL $\cap$ JQL				566 sources
SJL $\cap$ JQL				424 sources
MJL $\cap$ SJL $\cap$ JQL				393 sources
OpenAlex topic filtering				
MJL $\cap$ SJL $\cap$ JQL $\cap$ OA (OAS* long-list)				2,380 sources
OA sources with topics specifically in Economics or Commerce (OAS)				1,659 sources

We extract an initial corpus of works in the OAS source list, by retaining article and review types with publication years between 2020 and 2024 inclusive. We exclude works that are paratext or retracted, or without an authorship entity. The institution retention curve Fig. 1 implies that the preferred minimum cut

for works per institution per year could be chosen as $\tau_U = 10$, leaving approximately 1,900 institutions and 75% of the works. We will test values of $\tau_U = 5, 10$ and 15. For each work we compute the value of ω_{iu}

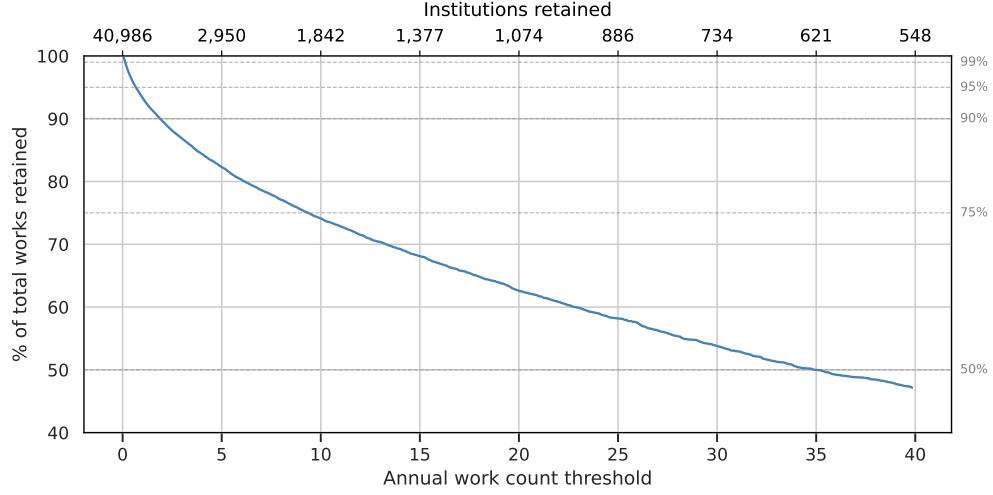


Figure 1: Institution retention curve

4 Results

Baseline model. [Placeholder.]

Parameter sensitivity. [Placeholder: $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$; $\chi \in \{0.00, 0.25, 0.50, 0.75, 1.00\}$; $\mathbf{m} \in \{(1, 0, 0, 0), (0, 1, 1, 0), (1, 1, 1, 1)\}$; $\rho \in \{0, 1\}$; τ sweep.]

Bootstraps. [Placeholder.]

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Prospects

[Placeholder.]

8 Addenda

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